

Disproof of conjecture 00000000408 at the staircase of order 2

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1 The claim

Conjecture 00000000408 reads:

Definition: Decomposition numbers $g(\lambda, \mu, \nu)$ (Kronecker coefficients of $s_\lambda * s_\mu$). Conjecture: For $\lambda = \mu = \nu$ of staircase shape, g is odd if and only if n is a triangular number (verifiable for $n \leq 20$).

The parenthetical identifies $g(\lambda, \mu, \nu)$ as the Kronecker coefficient of the internal product of Schur functions, equivalently the multiplicity

$$g(\lambda, \mu, \nu) = \langle \chi_\lambda \chi_\mu, \chi_\nu \rangle_{S_{|\lambda|}}.$$

The staircase of order $n \geq 1$ is the partition

$$\sigma_n = (n, n-1, \dots, 1)$$

of $N = n(n+1)/2$. The quantity in the conjecture is $g_n = g(\sigma_n, \sigma_n, \sigma_n)$, and a positive integer is triangular when it equals $k(k+1)/2$ for some integer $k \geq 0$. The claim is therefore

$$\forall n \geq 1, \quad g_n \text{ is odd} \iff n \text{ is triangular.}$$

This note disproves that equivalence at $n = 2$. The Lean development in `lean/Disproof.lean` checks the finite representation theory of S_3 and the arithmetic; it uses neither `sorry` nor `native.decide`.

2 The Specht module $S^{(2,1)}$

For $n = 2$ one has $\sigma_2 = (2, 1)$, a partition of 3. Let S_3 act on $\mathbb{Z}^3$ by permuting coordinates. The tabloid module of shape $(2, 1)$ has basis the three tabloids, indexed by the entry of the single box in the second row, and is this permutation representation. The polytabloids are the differences $e_i - e_j$. They span the sum-zero plane

$$S^{(2,1)} = \{(x, y, z) \in \mathbb{Z}^3 : x + y + z = 0\}.$$

That plane is free of rank 2 on the basis

$$b_1 = (1, -1, 0), \quad b_2 = (1, 0, -1).$$

Indeed, if $x + y + z = 0$, then

$$(x, y, z) = (-y) b_1 + (-z) b_2,$$

which is `reconstruct` in the Lean file. The trace of a permutation τ on $\mathbb{Z}^3$ is the number $\text{fix}(\tau)$ of fixed points. The invariant line spanned by $(1, 1, 1)$ contributes trace 1, so the trace on the complementary sum-zero plane is

$$\chi(\tau) = \text{fix}(\tau) - 1.$$

The same trace is recovered from the matrices of the six permutations in the basis (b_1, b_2) : the coordinate formula above gives the trace

$$\text{tr}(\tau) = -(\tau \cdot b_1)_y - (\tau \cdot b_2)_z,$$

and `trace_eq_chi` checks equality with χ on every element of S_3 . The six maps are exactly the bijections of $\{0, 1, 2\}$ (`group_exhaustive`).

The values are

class	identity	transpositions	3-cycles
size	1	3	2
χ	2	0	-1

3 Irreducibility

The only linear characters of S_3 are the trivial character and the sign character. A one-dimensional representation lands in $\{\pm 1\}$. Every 3-cycle has order 3, so it must act by 1. All transpositions are conjugate, so they share a common value ± 1 . The value $+1$ gives the trivial character and the value -1 gives the sign character.

Pairing the table above against these two characters gives

$$\begin{aligned} \sum_{\tau \in S_3} \chi(\tau) &= 2 + 3 \cdot 0 + 2 \cdot (-1) = 0, \\ \sum_{\tau \in S_3} \text{sgn}(\tau) \chi(\tau) &= 2 + 3 \cdot (-1) \cdot 0 + 2 \cdot (1) \cdot (-1) = 0, \\ \sum_{\tau \in S_3} \chi(\tau)^2 &= 2^2 + 3 \cdot 0^2 + 2 \cdot (-1)^2 = 6. \end{aligned}$$

Thus $\langle \chi, 1 \rangle = \langle \chi, \text{sgn} \rangle = 0$ and $\langle \chi, \chi \rangle = 1$ (`sumChi_eq`, `sumSignChi_eq`, `sumSquares_eq`, `characterInner_eq`). The group S_3 has three conjugacy classes, hence three irreducible characters. Their degrees d satisfy $d_1^2 + d_2^2 + d_3^2 = 6$. The two linear characters already contribute $1 + 1$, so the remaining irreducible character has degree 2. The character χ has degree 2, norm 1, and is orthogonal to both linear characters, so it is that remaining irreducible character. It is the character of the rank-2 module $S^{(2,1)}$ constructed above.

4 The Kronecker coefficient

By definition,

$$g(\sigma_2, \sigma_2, \sigma_2) = \frac{1}{|S_3|} \sum_{\tau \in S_3} \chi(\tau)^3 = \frac{1}{6} (2^3 + 3 \cdot 0^3 + 2 \cdot (-1)^3) = \frac{8-2}{6} = 1.$$

This is `sumCubes_eq` and `kronecker_eq`. In particular $g_2 = 1$ is odd (`kronecker_odd`).

5 Two is not triangular

Suppose $k(k+1) = 4$ for some integer $k \geq 0$. The products $k(k+1)$ for $k = 0, 1, 2$ are 0, 2, 6, and for $k \geq 3$ one has $k(k+1) \geq 12$. None equals 4. Hence 2 is not equal to $k(k+1)/2$ (`not_triangular_two`).

6 Conclusion

The integer g_2 is odd, while 2 is not a triangular number. The equivalence asserted for staircase order $n = 2$ is false, and therefore so is the universal claim. This is `staircase_two_breaks_the_equivalence`.